

INSPECTION CERTIFICATE EN 10204/3.1										CE22011590_3.1_01										CENA FITTINGS USA, LLC 400 N. SAM HOUSTON PARK EAST Houston (Texas), 77060 United States of America																			
Customer Order Y092C4A										Your Item Ref. Y1AB08:B.CI																													
Article/Specification Seamless elbows fully segmentable WPHY52 ASTM A860/A860M-18, MSS SP75-14, WPL6 ASTM A420/420M-20, ASME II Part.A-2021 SA420/SA420M, ASME B16.9/18, NACE MR0175/ISO15156-2015 NACE MR0103/ISO17945-2016, ASME II Part.D Tab.Y1																																							
Heat Treatment Hot formed at 780°C to 920°C - Normalized at 920°C,holding 1,5/min										Brand of manufacturer CENA																													
Marking Manufacturer - Od. x Th. - A860 WPHY52 - A/SA420 WPL6 - Heat Nr - OdxTh - SEGM																																							
Extent of material delivered Our pos. OY22001590/20000										Quantity NR 200										Article LR 90° 8" Std Painted HTS CODE 7307.93.30.40																			
Heat 923752										Marking code 923752										Certificate 01-22-15476 *										Supplier TENARIS-DALMINE									
Raw material Seamless pipe API 5L/ISO3183 X52 PSL2																																							
Results of chemical analysis %										<i>Ceq: C+ (Mn/6) + (Cr+Mo+V)/5 + (Cu+Ni)/15</i>																													
	C	Mn	Si	P	S	Cu	Ni	Cr	Mo	Al	Ti	Nb	V	N	B	Ceq	Pcm	Jfact.																					
min.		1,0000	0,1500							0,0200																													
max.	0,2000	1,4500	0,4000	0,0250	0,0100	0,3500	0,5000	0,2500	0,2500	0,0600	0,0400	0,0400	0,1000		0,00050	0,4200																							
Ladle	0,12	1,22	0,29	0,011	0,003	0,16	0,11	0,15	0,10	0,030	0,002	0,012	0,080	0,0088	0,0001	0,41																							
Check	0,13	1,21	0,29	0,011	0,003	0,16	0,11	0,15	0,09	0,032	0,002	0,010	0,070	0,0090	0,0001	0,41																							
Mechanical Tests: On fittings																																							
Specimen	Position	Direction	Temperature °C	Dimension mm	Yield Point N/mm²	Tensile Strength N/mm²	Elongation % Z"	Ys ----- Ts	Hardness HR 10% of batch 3 tests min.	Position	Direction	Temperature °C	Dimension mm	Impact Test - Specimen = KV																									
														Obtained energy				Shear Area																					
														Joule		% Values		Average		Values																			
24756.1.0	A	T	20	12,4x6,4	440	566	32,5	0,78	156-158	A	T	-50	10x5	55,0	70,0	82,0	69,0	60	65	65																			
The pipes are tested on tightness.																																							
Steel making process: Electric Furnace																																							
Specimen position: A=neutral axis of base material; W=weld; E=extrados of base material; I=intrados of base material; Z=heat affected zone																																							
Results of visual and dimensional inspection of fittings: SATISFACTORY Supplied products are in compliance with the requirements of the order The values of elongation and impact-test are according to PED 2014/68/EU The steel is fully killed and produced to a fine grain practice No weld repair, mercury and radiation free The material is in conformity with Arrête Ministeriel du 24 mars 1978 modifié																																							
24756.1.0: Rp = 63817 psi - Rm = 82091 psi																																							

Date 05/12/22	Inspection	Quality Control Manager BUTTURINI RICCARDO
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